fair-LDP: Uncertainty-Guided Fairness and Privacy for Federated Healthcare Learning

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2025 IEEE International Conference on Data Mining (ICDM) | 979-8-3315-9599-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDM65498.2025.00020

***Abstract*—Federated Learning (FL) offers a promising approach for collaborative model training in healthcare while preserving data privacy. However, existing FL methods often fall short in addressing two critical challenges: client-level fairness and compounded uncertainty from data heterogeneity and privacy-preserving mechanisms. We propose `fair-LDP`, a fairness-aware Local Differential Privacy framework that promotes fairness and privacy via uncertainty-guided aggregation in federated healthcare AI. `fair-LDP` leverages evidential neural networks (ENNs) to quantify predictive uncertainty and introduces a novel strategy that uses uncertainty-driven local differential privacy to guide fairness-aware updates while preserving data privacy. This ensures equitable performance across clients with varying data quality while mitigating the influence of unreliable or outlier updates. `fair-LDP` incorporates an adaptive mechanism that adjusts each client's privacy budget based on model performance, balancing fairness, privacy, and accuracy. We evaluate `fair-LDP` on real-world healthcare datasets under both IID and non-IID settings. Our experimental results show that it consistently outperforms state-of-the-art fairness-aware and privacy-preserving FL baselines, with no added computational overhead, while maintaining privacy guarantees comparable to homomorphic encryption and secure multiparty computation. By integrating uncertainty modeling, fairness-aware aggregation, and adaptive local differential privacy, `fair-LDP` provides a practical and principled solution for responsible, equitable, and privacy-preserving federated learning in healthcare.**

***Index Terms*—Smart Healthcare, Healthcare AI, Federated Learning, Ethical AI, Privacy, Fairness, Uncertainty in Machine Learning**

I. Introduction

Why Ethical Federated Learning Matters in Healthcare? Artificial intelligence (AI) has advanced modern healthcare, particularly in disease detection and management. In the case of Alzheimer's disease, early AI-enabled diagnosis is crucial for timely interventions that improve outcomes [1, 2]. However, demographic differences, imaging variations, and diverse clinical practices introduce data heterogeneity, hindering model generalizability [1, 2]. Federated learning (FL) offers a privacy-preserving solution by enabling collaborative model training across distributed devices [3]. Yet, ethical deployment of FL must address fairness and uncertainty. Ensuring client-level fairness is vital, as variations in data quality and quantity can lead to biased outcomes and reduced trust in AI systems [3, 4].

Why Uncertainty-Aware Fairness is Critical in Federated Healthcare AI? Uncertainty impacts AI performance in healthcare, especially in disease detection [5, 6]. In FL, non-IID and heterogeneous data amplify uncertainty, leading to inconsistent learning and unreliable predictions [7]. Uncertainty-aware methods support quality-aware aggregation by weighting client contributions based on reliability, reducing the impact of noisy or low-quality data [8]. Integrating uncertainty with fairness is essential for robust, equitable performance. We propose an uncertainty-aware fair FL framework that achieves high Alzheimer's detection accuracy while preserving client-level fairness across diverse datasets.

Why Balancing Privacy and Fairness Matters in Federated Healthcare AI? Differential privacy (DP) protects sensitive medical data by introducing noise [9, 10], but this can increase uncertainty and degrade performance for underrepresented groups. This trade-off poses risks to fairness and equity. To address it, we introduce `fair-LDP`, a fairness-aware local DP mechanism that adaptively allocates privacy budgets based on each client's performance. This approach balances privacy, fairness, and predictive accuracy, improving equity while preserving data confidentiality and model effectiveness.

Our work makes the following **key contributions**:

1) We propose `fair-LDP`, a novel fairness-aware local DP technique that adaptively allocates privacy budgets based on client performance. Using uncertainty estimates from evidential neural networks (ENNs), `fair-LDP` improves predictive accuracy for underrepresented groups while preserving strong privacy guarantees and system performance. To our knowledge, this is the first framework to integrate ENNs into FL for fairness-aware privacy control.
2) We enhance *client-level fairness* in FL by embedding fairness into the core of our uncertainty-aware framework. Unlike prior work that treats fairness as a secondary concern, we ensure consistent model performance across clients with diverse data quality and distributions [11, 12]. Our evaluation explicitly considers fairness threats, increasing the robustness and practical relevance of the system.
3) We present the first application of uncertainty-aware fair FL to Alzheimer's detection using real-world healthcare data [2, 13]. `fair-LDP` delivers state-of-the-art performance by jointly optimizing fairness, privacy, and accuracy,

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which has not been addressed in prior work.

4) We evaluate `fair-LDP` under two privacy threats, deep leakage and membership inference attacks, comparing it to three benchmark techniques. Our method matches their privacy and computational efficiency while outperforming them in both fairness and predictive performance.

II. Related work

A. Fairness-aware Federated Learning

Mohri et al. [14] proposed Agnostic Federated Learning (AFL), which enhances robustness and fairness by minimizing worst-case loss across heterogeneous client distributions. Li et al. [15] introduced q-Fair Federated Learning (q-FFL), prioritizing high-loss clients to reduce performance disparities in non-IID settings. q-FFL includes three variants: q-FedAvg (loss-weighted FedAvg), q-FedSGD (fairness-aware SGD), and q-MAML (fair meta-learning for client generalization). Moreover, Li et al. [16] developed Ditto, a personalized FL framework that jointly optimizes global and local models but lacks explicit fairness evaluation. Ezzeldin et al. [11] proposed FairFed, which adjusts aggregation weights based on fairness metrics computed on local datasets, while preserving privacy via secure aggregation. To mitigate inter-institutional disparities, Hosseini et al. [12] proposed Prop-FFL, which minimizes performance variation among hospitals.

While existing work addresses collaborative and group fairness, applications in real-world healthcare remain limited. Notably, Li et al. [15] considers group fairness but does not evaluate it on healthcare data. Our work addresses this gap by introducing an uncertainty-aware fairness framework tailored for healthcare, focusing on Alzheimer's disease, and advancing client-level fairness beyond prior methods.

B. FL-based Disease Diagnosis and Detection

Meerza et al. [2] introduced the first FL approach for Alzheimer's diagnosis using spontaneous speech, employing q-FedAvg and q-FedSGD to reduce bias from data heterogeneity. Zhang et al. [17] applied homomorphic encryption to protect local models in FL-based IoT healthcare. Li et al. [18] developed ADDetector, a privacy-preserving Alzheimer's detection system using asynchronous FL and linguistic features from smart speakers. Moreover, Khalil et al. [13] designed an efficient FL-based AD prediction model with hardware acceleration and local data privacy. Beyond Alzheimer's, Kumar et al. [19] and Singh et al. [20] applied FL with blockchain for COVID-19 detection and smart healthcare monitoring, respectively.

While several studies address fairness, few systematically evaluate it under adversarial conditions, an essential aspect for real-world deployment. We address this gap by analyzing fairness in the presence of attacks and evaluating performance on real-world healthcare datasets using formal fairness metrics.

C. Uncertainty-Aware AI for Disease Diagnosis

MacDonald et al. [6] emphasized the role of uncertainty in AI-driven healthcare for improving decision-making, safety, and fairness, especially for underrepresented groups. Tabarisaadi et al. [21] enhanced skin cancer detection using uncertainty quantification (UQ) methods such as MC Dropout, Bayesian Ensembling, and SNGP. Ghoshal et al. [22] applied Bayesian CNNs to flag unreliable predictions in pancreatic adenocarcinoma grading, while Prince et al. [23] used Bayesian deep learning to detect uncertainty in ACP diagnosis from MRI. Wang et al. [24] proposed FedUAA, an FL model for diabetic retinopathy that adjusts client aggregation based on uncertainty-aware weighting.

While UQ improves reliability, calibration often varies across demographic groups, leading to uneven confidence estimates. Most prior work also overlooks security threats such as adversarial attacks. Our approach addresses both challenges by leveraging uncertainty-aware techniques to improve fairness and robustness in federated healthcare systems under adversarial conditions.

D. Privacy-Preserving FL-based Healthcare Systems

Zhang et al. [17] used cryptographic techniques in IoT and FL-based healthcare to protect local models against attacks such as model reconstruction and inversion. Yazdinejad et al. [25] introduced AP2FL, a privacy-preserving FL model using Trusted Execution Environments (TEEs) to secure clients and servers during training. They combined Active Personalized FL (ActPerFL) and Batch Normalization to mitigate performance loss from data heterogeneity. Additionally, Islam et al. [26] implemented a differential privacy-enabled FL framework with feature selection to preserve data privacy while reducing dimensionality and computational cost. Kumar et al. [19] developed a blockchain-based FL model for COVID-19 detection using CT images, applying normalization to handle inter-hospital data heterogeneity.

Although these works claim to preserve privacy, many do not evaluate resilience against privacy attacks or report privacy metrics, limiting the validation of their approaches. Moreover, the interplay between privacy and fairness remains largely unexplored. Our work addresses these gaps by proposing a fairness-aware, privacy-preserving FL framework and evaluating its robustness against both privacy and fairness threats.

III. Preliminaries

Our proposed `fair-LDP` framework leverages Evidential Neural Networks (ENNs) [27] for uncertainty estimation and incorporates Local Differential Privacy (LDP) [28] to ensure data protection, as detailed below.

A. Evidential Neural Networks (ENNs)

An Evidential Neural Network (ENN) is similar to traditional neural networks but replaces the softmax layer with an activation function, such as ReLU, to ensure non-negative outputs. These outputs form an evidence vector, which models a predicted Dirichlet distribution [27] based on *Dempster-Shafer theory* [29] and *Subjective Logic* [30]. Rather than providing a single class label, the ENN generates a belief function to quantify the confidence in different outcomes.

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Building on Sensoy et al. [27], Hu et al. [31] improved the approach by directly measuring the quality of the predicted Dirichlet distributions. For a given sample i, the network predicts an evidence vector $f(\mathbf{x}_i|\Theta)$, where $\mathbf{x}_i \in \mathbb{R}^L$ is the input and Θ are the network parameters. The associated Dirichlet distribution has parameters $\boldsymbol{\alpha}_i = f(\mathbf{x}_i|\Theta) + 1$, and the Dirichlet density $\text{Dir}(\mathbf{p}_i; \boldsymbol{\alpha})$ serves as the prior for the Multinomial distribution $\text{Multi}(y_i|\mathbf{p}_i)$. To estimate the parameters $\boldsymbol{\alpha}_i$, a sum of squared loss function is formulated based on the Dirichlet PDF:

$$\mathcal{L}(f(\mathbf{x}_i|\Theta), y_i) = \sum_{j=1}^{K} \left(y_{ij}^2 - 2y_{ij}\mathbb{E}[p_{ij}] + \mathbb{E}[p_{ij}^2] \right),$$

where $\mathbb{E}$ denotes the expected probability distribution.

B. Differential Privacy

Definition 1 [28]: (ϵ, δ)-differential privacy is a relaxation of standard ϵ-differential privacy that permits a small probability of failure, denoted by δ. Let $\mathcal{D}$ be the set of all possible datasets, and $\mathcal{M} : \mathcal{D} \to \mathcal{O}$ be a randomized algorithm that outputs an element from the output space $\mathcal{O}$. The algorithm $\mathcal{M}$ satisfies (ϵ, δ)-differential privacy if, for all neighboring datasets D_1 and D_2 in $\mathcal{D}$, and for all outputs $O \in \mathcal{O}$, the following holds:

$$\Pr[\mathcal{M}(D_1) = O] \leq e^{\epsilon} \cdot \Pr[\mathcal{M}(D_2) = O] + \delta \qquad (1)$$

Here, $\epsilon \geq 0$ is the privacy parameter that quantifies the strength of the privacy guarantee, with smaller values indicating stronger privacy.

In the context of FL, two variants of differential privacy are commonly used. Local Differential Privacy (LDP) [32] requires each participant to add noise to their updates before sending them to the server, ensuring privacy without trusting the server. In contrast, Central Differential Privacy (CDP) [33] relies on the server to apply a DP mechanism during aggregation. Our approach adopts LDP, eliminating the need to trust the central server.

IV. Problem Statement

This study employs a framework for FL considering uncertainty in detecting Alzheimer's disease while prioritizing fairness at the client level. The system is designed as a horizontal FL model, with a central global model stored on a cloud server and local models running at different hospitals (i.e., clients). This decentralized method preserves data privacy, as each hospital trains its own model with patient data and shares periodic updates with the global model. The primary challenge lies in ensuring equitable updates from local models and mitigating data variability across hospitals.

Another key challenge arises from integrating differential privacy (DP), which, while essential for ensuring patient confidentiality and regulatory compliance, introduces noise into model updates that can degrade accuracy. This degradation is often uneven across clients, disproportionately affecting institutions with smaller or more heterogeneous datasets and raising concerns about fairness. Thus, balancing strong privacy guarantees with model performance and client-level fairness remains a critical objective of this study.

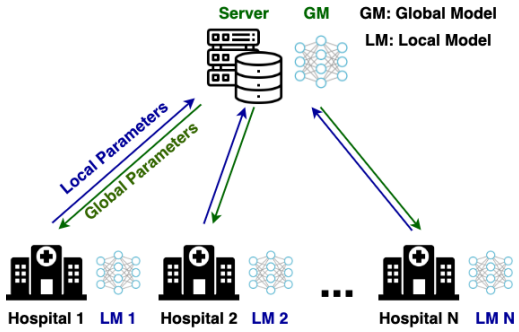


Fig. 1. Network Model of a Federated Learning Healthcare System

This work introduces `fair-LDP`, a novel algorithm that enhances *fairness* in FL while preserving the same local differential privacy (LDP) guarantees as baseline methods. `fair-LDP` significantly improves both prediction accuracy and fairness by incorporating a *fairness-aware perturbation mechanism* that strategically calibrates LDP noise. This ensures client updates remain private and equitable, enabling better adaptation to data heterogeneity in sensitive healthcare applications. The optimization objective is formalized as:

$$\begin{aligned} \text{maximize} \quad & \mathcal{ACC}(M(\theta^* + \eta)) + CF(M(\theta^* + \eta)), \\ \text{subject to} \quad & \epsilon \leq \epsilon'. \end{aligned} \qquad (2)$$

Here, $\eta \sim \mathcal{N}(0, \sigma^2)$ is Gaussian noise with $\sigma \propto 1/\epsilon$, where ϵ denotes the privacy budget under `fair-LDP`, and ϵ' is the LDP budget enforced by the baseline. Lower ϵ implies stronger privacy, but at the expense of reduced utility (e.g., lower prediction accuracy). $\mathcal{ACC}(M(\theta^+\eta))$ represents the prediction accuracy of local models using DP-applied parameters $\theta^+\eta$, while $CF(M(\theta^* + \eta))$ denotes client-level fairness, measured by the fairness metrics detailed in Section VII-D.

V. System Model

A. Network Model

The smart healthcare system adopts a network model (Fig. 1) connecting multiple hospitals (clients) to a central cloud server using horizontal federated learning (HFL), where data share the same feature space (e.g., MRI images) but differ in sample space (patients). Each hospital trains a local model on its patient data and sends updates to the server, which aggregates them to improve a shared deep learning model for Alzheimer's disease detection. The server redistributes the updated model, and this cycle repeats until convergence. This framework enables collaborative model training across institutions while preserving patient privacy and data security.

B. Node Model

In the healthcare network, individual hospitals act as clients, each maintaining a local dataset comprising patient records, medical histories, imaging data, and so forth. To comply with

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data protection laws and safeguard patient privacy, hospitals do not share raw data. Instead, each hospital trains a local model using its own data and computes updates (e.g., gradients), which are then sent to a central server for aggregation. This decentralized approach enables collaborative model development while keeping patient data secure. However, due to differences in demographics, disease prevalence, and medical practices across hospitals, achieving a generalized global model remains challenging. The central server serves as the aggregator and coordinator in the federated learning process, never accessing raw data. It collects updates from all participating hospitals, aggregates them into a global model, and redistributes the updated model for continued local training. This iterative process continues over multiple communication rounds until the model converges. The server plays a critical role in maintaining privacy while harnessing data diversity to build robust and collaborative predictive models.

C. Threat Model

To evaluate vulnerabilities in FL systems and the effectiveness of `fair-LDP`, we consider adversarial behaviors by compromised clients aiming to disrupt training by modifying local parameters or data, termed *fairness attacks*, as such actions can amplify performance disparities and violate fairness guarantees of consistent local performance [34].

- ***Byzantine Attacks*** [35]: These attacks inject arbitrary metrics via tampered SGD updates, often targeting clients (e.g., hospitals) to prolong training or cause model divergence. We apply Gaussian noise to perturb model parameters of compromised clients before updates are sent to the server. Such attacks can skew the global model, degrade performance on certain data distributions, and disproportionately affect clients whose data aligns with the poisoned updates, causing unfair outcomes.
- ***Poisoning Attacks*** [36]: These compromise client datasets by subtly altering inputs to mislead model predictions. We adopt a backdoor-based approach [37], where a malicious client embeds a trigger in its local data to manipulate the global model. The model performs normally on benign inputs but outputs attacker-defined labels when the trigger appears, leading to biased learning and harming clients with overlapping classes, thus violating fairness.

To assess `fair-LDP`'s privacy-preserving effectiveness and its impact on fairness and accuracy, we further examine:

- **Deep Leakage from Gradients (DLG)** [38]: DLG reconstructs training data by minimizing the difference between shared gradients and dummy gradients derived from synthetic inputs, eventually recovering original inputs and labels without auxiliary information.
- **Membership Inference Attacks (MIAs)** [39]: MIAs test whether a specific data point was part of the training set by analyzing the model's outputs, revealing individual record membership and exposing privacy vulnerabilities.

Algorithm 1 Fairness-Aware LDP (`fair-LDP`)

Require: Total number of clients N, number of selected clients K, total communication rounds T, local training epochs E, client dataset $\mathcal{D}_k$, initial global model parameters θ_0, total DP budget ϵ_{total}, small constant α to avoid zero allocation

for $t = 0$ to $T-1$ **do**
  Server randomly selects K clients
  Server sends current global model θ_t to selected clients
  for each selected client k **do**
    Perform local training for E epochs on $\mathcal{D}_k$ to obtain θ_t^k
    Compute model uncertainty u_k and send to server
  end for
  Server computes mean uncertainty $\bar{u} = \frac{1}{K}\sum_{k=1}^{K} u_k$ and broadcasts it to selected clients
  for each selected client k **do**
    Compute fairness score: $S_k = |u_k - \bar{u}| + \alpha$
    Normalize score: $w_k = \frac{S_k}{\sum_{j=1}^{K} S_j}$
    Assign personalized privacy budget: $\epsilon_k = w_k \cdot \epsilon_{\text{total}}$
    Compute gradients $\nabla_{\theta_t^k}$ on $(\mathcal{D}_k, \theta_t^k)$
    Add Gaussian noise: $\nabla_{\theta_t^k}^{DP} = \nabla_{\theta_t^k} + \mathcal{N}(0, \sigma_k^2)$, where $\sigma_k^2 = \frac{1}{\epsilon_k^2}$
    Update the local model: $\theta_t^k \leftarrow \theta_t^k - \eta \cdot \nabla_{\theta_t^k}^{DP}$
  end for
  Server aggregates local models to update the global model:

$$\theta_{t+1} = \frac{1}{K}\sum_{k=1}^{K} \theta_t^k$$

end for
Output: Final global model θ_T

VI. Proposed Approach: FAIR-LDP

In FL, ***client-level fairness*** emphasizes uniform performance across clients, typically measured by the variance in local model accuracies [15].

Definition 2 [15]: A solution w is fairer than w' if

$$\mathbf{Var}(F_1(w), \ldots, F_m(w)) < \mathbf{Var}(F_1(w'), \ldots, F_m(w')), \quad (3)$$

where $F_i(w)$ denotes the performance of the i-th client.

Ensuring fairness in smart healthcare is crucial to providing equitable diagnoses across hospitals. Inconsistent performance exacerbates disparities, while uniform performance enhances trust and supports reliable care across diverse populations.

Traditional Local Differential Privacy (LDP) methods add identical noise to all clients, often disproportionately degrading performance for underrepresented or data-scarce clients. To address this, we propose `fair-LDP`, a *fairness-aware local differential privacy mechanism* that adaptively allocates privacy budgets based on each client's performance.

`fair-LDP` extends the standard FedAvg protocol by integrating fairness-aware DP adjustments. In each communication round, the server selects clients, sends them the global

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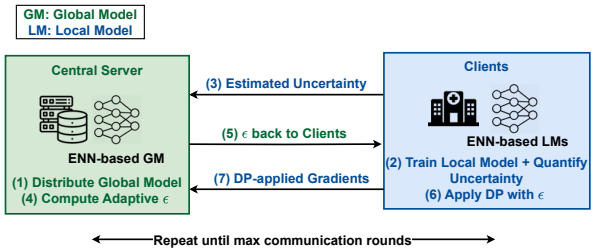


Fig. 2. Process of `fair-LDP`

model, and collects updated models after they have completed local training. Clients also send their local model uncertainty, and the server computes the global mean uncertainty. Based on this, the server calculates a fairness score S_k for each client using its uncertainty deviation from the mean, where uncertainty is quantified via Evidential Neural Networks (ENNs) [27] using Subjective Logic (SL) [30]. SL provides belief-based outputs that reflect both evidence and uncertainty, offering a robust signal across heterogeneous data distributions.

Unlike accuracy or loss, uncertainty captures the confidence in a prediction, making it more suitable for fairness assessment. The fairness score S_k is then used to determine the client's share of the total DP budget: $w_k = \frac{S_k}{\sum_{i=1}^{K} S_i}$, and $\epsilon_k = w_k \cdot \epsilon_{total}$. A higher S_k (i.e., greater deviation from the mean) results in a larger ϵ_k, meaning less noise and better accuracy for clients most affected by fairness gaps.

Clients perturb their gradients using these personalized privacy budgets, and the server aggregates the noisy updates to form the new global model. This design enhances fairness while preserving competitive accuracy and privacy guarantees, enabling balanced performance across diverse clients. The above process is repeated iteratively until the specified number of communication rounds is completed, as shown in Fig. 2.

VII. Experimental Setup

A. Datasets

We evaluate our proposed approach using one benchmark dataset and two real-world datasets:

- **MNIST** [40] is a standard benchmark comprising 70,000 grayscale images of handwritten digits (0–9), each 28×28 pixels. It includes 60,000 training and 10,000 test images, commonly used for evaluating image classification and pattern recognition models.
- **Alzheimer's Disease Neuroimaging Initiative (ADNI)** [41] is a large-scale, longitudinal study launched in 2004 to advance Alzheimer's research. It provides 3D volumetric MRI scans for multi-class classification of Alzheimer's disease (AD), mild cognitive impairment (MCI), and cognitively normal (CN) subjects. The preprocessed tabular dataset includes 10,494 samples with 87 neuroimaging-derived features.
- **Alzheimer MRI (AD-MRI)** [42] contains MRI brain images across 32 horizontal slices, categorized into four dementia severity levels: Mild, Moderate, Very Mild, and Non-Dementia. The dataset is balanced via oversampling, with 3,200 samples per class.

B. Parameterization

The hyperparameters are carefully selected to optimize the performance of the federated learning framework. We use a learning rate of 0.001 and a regularization coefficient (λ) of 0.001 to balance loss components. Each client performs local training with a batch size of 64 for 20 epochs per round. The framework runs for 40 communication rounds with five clients selected per round. To ensure robustness, each experiment is repeated 10 times. An evidential neural network (ENN) is employed due to its effectiveness in classification and uncertainty quantification, which follows the architecture of a convolutional neural network (CNN) by replacing the last layer from Softmax to ReLU. For the MNIST dataset, the CNN comprises two convolutional layers (each with 32 and 64 filters of size 5×5, followed by max-pooling), a flattening layer, and a fully connected layer with 1024 ReLU units. For the ADNI dataset, the model is a fully connected neural network with three dense layers (512, 256, and 128 ReLU units), followed by a 50% dropout layer. For the AD-MRI dataset, the model consists of three convolutional layers (32 filters of 5×5, 64 filters of 5×5, and 128 filters of 3×3), each followed by max-pooling. The features are flattened and passed through a fully connected layer with 1024 ReLU units and a 50% dropout. All models have a final dense layer producing the class logits. The dataset on each client is randomly split into 80% for training and 20% for testing. All experiments are conducted on Virginia Tech's Advanced Research Computing (ARC) infrastructure, utilizing both the Falcon and TinkerCliffs high-performance computing clusters. Falcon nodes feature 4×NVIDIA A30 GPUs (24 GB), dual AMD EPYC 7543 CPUs, and 512 GB RAM. TinkerCliffs A100 nodes have 8×NVIDIA A100 GPUs (80 GB), dual AMD EPYC 7742 CPUs, and 2 TB RAM. Table I summarizes the key parameters and their default values.

For privacy-preserving mechanisms, we choose parameters to achieve a good balance between privacy level and either accuracy or efficiency, depending on the technique used. For Local Differential Privacy (LDP), we use an epsilon value of 1.0 and a delta value of $1e^{-5}$ to define the privacy budget and the acceptable risk of privacy leakage. The sensitivity is set to 1.0 to determine the maximum influence of any single data point, and the noise is added using Gaussian mechanisms. In Homomorphic Encryption (HE), we set the polynomial modulus degree to 4096 and specify the coefficient modulus bit sizes as a list of [40, 20, 40] to manage security level and computational cost. The global scale is set to 2 to the power of 40 for encoding precision in CKKS. For Secure Multi-Party Computation (SMC), we use a threshold of 7 to define the minimum number of shares required to reconstruct the secret and distribute the data among 10 total parties.

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TABLE I
KEY DESIGN PARAMETERS AND DEFAULT VALUES

Parameter	Value
Total number of clients	10
Communication rounds	40
Local training epochs	20
Simulation runs	10
Learning rate	0.001
Loss balance coefficient (λ)	0.001
Batch size	64

C. Comparing Schemes

To evaluate the effectiveness and efficiency of the proposed `fair-LDP`, we compare it against the following FL schemes categorized by their support for fairness and privacy:

- **Fair FL without Privacy**: This baseline emphasizes client-level fairness without applying any privacy-preserving mechanisms (e.g., DP or secure aggregation). We consider:
  - **Agnostic Federated Learning (AFL)** [14] optimizes performance across all possible mixtures of client data, rather than the average, to enhance robustness under distributional shifts.
  - **q-Fair Federated Learning (q-FFL)** [15] improves fairness by prioritizing clients with higher loss. The tunable parameter q adjusts the trade-off between global accuracy and fairness across heterogeneous clients.
- **Privacy-Preserving FL without Fairness**: This category includes schemes that prioritize privacy, but not fairness:
  - **Local Differential Privacy (LDP)** [28] adds noise to gradients to protect individual data, improving privacy but potentially harming utility under strong noise.
  - **Homomorphic Encryption (HE)** [43] enables computation on encrypted data, ensuring confidentiality but incurring high computational and communication overhead.
  - **Secure Multiparty Computation (SMC)** [44] allows clients to train collaboratively by exchanging secret-shared updates. While offering strong privacy, SMC faces scalability and efficiency challenges due to its cryptographic complexity.
- **Fair and Privacy-Preserving FL**: To benchmark against methods that address both fairness and privacy, we adopt a modular combination approach. Since most existing methods define fairness from a centralized group fairness perspective (e.g., across sensitive attributes) rather than client-level fairness, we construct a hybrid baseline by combining q-FFL with LDP. This pairing aligns with our `fair-LDP` design, enabling a consistent and meaningful comparison focused on client-level fairness under privacy constraints.

D. Metrics

We evaluate the performance of `fair-LDP` using prediction accuracy, fairness, privacy, and efficiency metrics:

- **Prediction Accuracy (Acc)** measures the agreement between model predictions and ground truth labels for Alzheimer's disease detection, computed as the test accuracy at each communication round.
- **Loss Disparity (LD)** [45] captures the variance in test loss across clients:

$$LD = \frac{1}{N}\sum_{i=1}^{N} \mathrm{Var}(\mathcal{L}_i), \tag{4}$$

where $\mathcal{L}_i$ is the test loss for client i, and N is the number of clients. Lower LD indicates more balanced model performance.
- **Accuracy Disparity (AD)** [15] quantifies the variance in accuracy across clients:

$$AD = \frac{1}{N}\sum_{i=1}^{N} \left(\mathrm{Accuracy}_i - \overline{\mathrm{Accuracy}}\right)^2, \tag{5}$$

where $\mathrm{Accuracy}_i$ is the test accuracy for client i, and $\overline{\mathrm{Accuracy}}$ is the mean accuracy across clients. Lower AD implies better fairness in predictive performance.
- **Gradient Match Loss (GML)** [38] evaluates the deviation between true and perturbed gradients after applying privacy mechanisms:

$$\mathrm{GML} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{M_i} |\mathrm{grad}_{i,j}^{\mathrm{real}} - \mathrm{grad}_{i,j}^{\mathrm{reconstructed}}|}{\sum_{i=1}^{N} M_i}, \tag{6}$$

where M_i is the number of gradient elements in layer i, and N is the total number of layers. A higher GML value indicates a stronger level of privacy preservation, as it reflects a greater deviation between the true and reconstructed gradients.
- **Membership Inference Attack Success Rate (ASR)** [46] measures the success rate of identifying whether a sample was in the training set:

$$\mathrm{ASR} = \frac{\sum_{j=1}^{m} \mathbf{1}\left(C(x_j) = M_S(x_j)\right)}{m}, \tag{7}$$

where $C(x_j)$ is the adversary's prediction, $M_S(x_j)$ is the true membership status, and $\mathbf{1}(\cdot)$ is the indicator function. Lower ASR indicates stronger privacy.
- **Run Time (RT)** measures the execution time of privacy-preserving algorithms to assess computational efficiency.

VIII. NUMERICAL RESULTS & ANALYSIS

In this section, we present a comprehensive evaluation of the proposed `fair-LDP` framework by comparing its performance against state-of-the-art baseline methods under fairness threat scenarios across multiple datasets and metrics.

A. Comparison Analyses

Tables II and III provide an in-depth comparison of the performance of state-of-the-art schemes for evaluating fairness performance across benchmark and healthcare datasets, including MNIST, ADNI, and AD-MRI. These results are generated under fairness threat scenarios described in Section V-C, where datasets are deliberately compromised to simulate real-world bias and data heterogeneity. For q-FFL, we set the fairness sensitivity parameter $q = 0.0001$ across all experiments to ensure consistency and fairness in comparison,

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TABLE II
SUMMARY OF PERFORMANCE COMPARISON UNDER **FAIRNESS THREATS** (PART 1)

Data Type	PP FL without Fairness			Fair and PP FL			Our Approach		
	LDP			q-FFL + LDP			`fair-LDP`		
	Acc	AV	LD	Acc	AV	LD	Acc	AV	LD
MNIST									
IID	90.540	25.170	0.0297	81.327	70.353	0.0341	**91.595**	**18.339**	**0.0284**
Non-IID	58.790	359.015	0.1336	**65.473**	**260.181**	**0.0605**	59.184	345.009	0.1405
ADNI									
IID	73.727	8.255	0.0155	66.912	7.587	**0.0077**	**74.542**	**5.224**	0.0146
Non-IID	59.199	86.794	0.1428	58.273	72.273	**0.0328**	**59.770**	**59.894**	0.1180
AD-MRI									
IID	72.961	9.257	6.4936	≤ 50.0	**5.887**	**0.0001**	**75.867**	7.006	1.4547
Non-IID	75.141	182.966	0.4398	≤ 50.0	704.667	**0.0022**	**78.251**	**135.297**	0.2196

TABLE III
SUMMARY OF PERFORMANCE COMPARISON UNDER **FAIRNESS THREATS** (PART 2)

Data Type	Fair FL without PP						PP FL without Fairness					
	Ditto			q-FFL			SMC			HE		
	Acc	AV	LD	Acc	AV	LD	Acc	AV	LD	Acc	AV	LD
MNIST												
IID	55.074	230.321	0.1790	92.011	22.714	0.0191	92.498	20.528	0.0183	92.247	21.419	0.0174
Non-IID	77.643	133.695	0.1875	72.798	277.510	0.0899	71.169	321.392	0.1080	71.544	314.974	0.1209
ADNI												
IID	51.084	2.340	0.0138	73.782	4.450	0.0016	72.704	7.703	0.0024	73.992	6.370	0.0024
Non-IID	≤ 50.0	72.114	0.0306	65.735	72.161	0.0087	65.195	77.528	0.0092	64.683	83.053	0.0086
AD-MRI												
IID	65.215	13.038	0.0006	90.539	3.404	0.0159	90.484	4.065	0.0161	90.438	2.433	0.0158
Non-IID	92.983	17.481	0.0052	81.835	129.512	0.0453	81.812	139.357	0.0482	81.878	123.941	0.0425

while also aiming to preserve a reasonable level of prediction accuracy. Our proposed approach, `fair-LDP`, consistently outperforms baseline methods by achieving a strong balance between accuracy and fairness under both IID and non-IID settings. Compared to traditional privacy-preserving methods like LDP, our method significantly improves fairness, reducing loss disparity by up to 75% on the AD-MRI dataset (IID), while maintaining or even improving accuracy. On the ADNI dataset (IID), `fair-LDP` achieves the highest accuracy and reduces LD by over 40% compared to combined fairness-privacy baselines.

One observation with the q-FFL + LDP scheme is that since q-FFL employs loss-based aggregation, giving more weight to clients with higher local losses, it performs relatively well in terms of loss disparity (LD), especially on ADNI datasets. However, it shows weaker performance on accuracy variance (AV), highlighting an inconsistency between different fairness metrics. This reveals that optimizing for one aspect of fairness does not guarantee fairness across other dimensions. Moreover, the outperformance of q-FFL + LDP on the MNIST non-IID dataset suggests it may be more effective on simpler benchmark datasets, where the data complexity and distribution are less challenging. However, this advantage does not generalize to more complex settings: q-FFL + LDP suffers from a significant drop in prediction accuracy, falling below 50% in cases like the AD-MRI dataset, where the model fails to maintain predictive performance under a privacy constraint. The modest improvement in LD comes at the cost of severely degraded accuracy, limiting the method's practicality. We also observe that Ditto, a personalization-based approach, performs well under non-IID settings by tailoring models to individual clients, mitigating data heterogeneity. However, in IID settings, it degrades performance, showing lower accuracy and higher disparity. This is because personalized updates unnecessarily diverge from a well-generalized global model, introducing noise where a single shared model would be more effective. In contrast, even in more challenging non-IID scenarios, `fair-LDP` remains robust, improving fairness by 15–50% over privacy-only or fairness-only baselines, while avoiding the steep accuracy loss observed in methods like q-FFL + LDP. These results validate that `fair-LDP` successfully maintains high prediction accuracy while substantially improving fairness, making it a practical and scalable solution for fairness-sensitive FL systems in healthcare and other high-stakes domains.

For the ADNI and AD-MRI datasets, although overall

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TABLE IV
SUMMARY OF COMPARISON OF PRIVACY PRESERVATION UNDER **PRIVACY THREATS** FOR `fair-LDP`

Data Type	PP FL without Fairness									Fair and PP FL			Our Approach		
	SMC			HE			LDP			q-FFL + LDP			`fair-LDP`		
	GML ($\times 10^7$)	ASR	RT	GML ($\times 10^7$)	ASR	RT	GML ($\times 10^7$)	ASR	RT	GML ($\times 10^7$)	ASR	RT	GML ($\times 10^7$)	ASR	RT
MNIST															
IID	2.955	~ 0	13.088	N/A	~ 0	166.620	1.266	$\sim$ **0**	6.392	1.266	~ 0	5.935	**1.303**	0.0001	**5.172**
Non-IID	2.964	~ 0	13.137	N/A	~ 0	168.203	1.266	0.0011	5.601	1.266	~ 0	6.118	**1.393**	$\sim$ **0**	**4.662**
ADNI															
IID	0.1951	~ 0	5.771	N/A	~ 0	130.003	0.0810	0.3337	2.179	0.0810	0.5521	2.143	**0.0823**	**0.1155**	2.163
Non-IID	0.1915	~ 0	5.894	N/A	~ 0	137.933	0.0811	0.5532	2.146	0.0812	0.4338	2.131	**0.0872**	**0.3999**	2.161
AD-MRI															
IID	92.605	~ 0	323.996	N/A	~ 0	150.003	39.773	~ 0	**69.948**	39.773	~ 0	70.740	**40.831**	$\sim$ **0**	70.053
Non-IID	92.605	~ 0	234.953	N/A	~ 0	159.009	39.773	~ 0	69.984	39.773	~ 0	70.027	**44.067**	$\sim$ **0**	**69.934**

accuracy is lower than MNIST, this is primarily due to the limited number of samples and severe class imbalance inherent in these medical datasets. We apply data oversampling to ensure each client has sufficient data, but class imbalance still negatively impacts predictive performance. Despite this challenge, `fair-LDP` consistently outperforms baseline methods in both accuracy and fairness, showing its robustness. We believe that with access to larger and more balanced datasets, our method has the generalizability to achieve even stronger performance, particularly in real-world healthcare applications where fairness and privacy are both critical.

The modest underperformance of `fair-LDP` in accuracy relative to HE and SMC stems from the fact that these cryptographic methods preserve data utility more fully by enabling exact computations on encrypted data. However, this comes at a significant cost: HE and SMC are computationally expensive, resource-intensive, and difficult to scale, especially in edge-device or large-scale FL settings. In contrast, `fair-LDP` offers a lightweight, scalable solution with built-in fairness, making it a more practical and ethical alternative for real-world deployments.

Table IV presents a comprehensive comparison of state-of-the-art methods based on privacy preservation metrics across the MNIST, ADNI, and AD-MRI datasets. Note that we exclude fairness-only baselines from this comparison, as privacy metrics such as Gradient Matching Loss (GML) measure the discrepancy between gradients before and after applying privacy-preserving techniques. In the absence of any PP mechanism, this discrepancy is inherently zero, corresponding to the worst possible privacy performance. This implies that an attacker could potentially reconstruct the complete original gradients and thereby infer sensitive data, through deep leakage techniques, as described in Section V-C. Therefore, only methods with PP components are considered in this table. Our proposed method, `fair-LDP`, shows strong privacy protection with competitive GML and runtime, while maintaining high resilience to privacy attacks. Cryptographic methods like HE and SMC achieve near-zero ASR due to their encryption mechanisms. In HE, gradients are fully encrypted, making it impossible to compute GML, hence marked as N/A. In contrast, while SMC also applies encryption, it uses additive secure aggregation, which retains enough numerical structure to approximate gradient differences, allowing GML computation. Despite these advantages, both HE and SMC incur substantially higher runtime and are less practical for scalable, real-world deployments.

Our approach, `fair-LDP`, maintains a similarly low ASR to LDP and q-FFL + LDP, despite being a more lightweight and scalable solution. Notably, on the ADNI dataset, all LDP-based methods, including ours, exhibit slightly higher ASR values. This suggests that tabular data, such as ADNI, is generally more vulnerable to MIA attacks than image datasets like MNIST or AD-MRI. Additionally, ASR values are effectively zero on the AD-MRI dataset, indicating that complex, high-dimensional data, such as MRI images, are inherently more resistant to MIA than simpler image types (e.g., handwritten digits). Another consistent observation is that higher GML generally corresponds to lower ASR, as seen with SMC and our method on certain datasets. This reflects the expected trade-off where greater gradient distortion leads to reduced information leakage, an important indicator that the GML metric correlates well with actual privacy performance.

Compared to LDP and q-FFL + LDP, `fair-LDP` achieves similar or better GML with faster runtime, while offering a more balanced and scalable solution. More importantly, when combined with the results in Tables II and III, our method not only preserves privacy and prediction accuracy but also shows substantial improvements in fairness. This demonstrates that `fair-LDP` effectively balances accuracy, privacy, and fairness, making it a robust and practical solution for FL in sensitive domains like healthcare.

IX. CONCLUSION & FUTURE WORK

This work presents `fair-LDP`, an uncertainty-guided, fairness-aware local differential privacy framework that enhances prediction, fairness, and privacy in AI models for healthcare applications, such as Alzheimer's disease detection. `fair-LDP` consistently achieves better fairness than standard

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LDP while maintaining similar accuracy and variance on real-world healthcare datasets. Notably, it introduces no additional computational overhead compared to standard LDP and is significantly more efficient than encryption-based methods like HE and SMC, which rely on costly cryptographic operations. Although `fair-LDP` slightly underperforms HE and SMC in fairness and accuracy, due to their access to decrypted model gradients during evaluation, it offers a strong trade-off across accuracy, fairness, privacy, and efficiency, making it a practical choice when fairness is a key priority.

For the **future work**, we aim to leverage explainable AI techniques into current approaches to improve the interpretability of the models and provide transparent reasons for predictions. This is particularly important in healthcare applications, where clinicians and patients need to trust the predictions. Furthermore, we aim to extend our work to incorporate multi-modal data, which will improve the performance and generalization of predictive models in healthcare contexts. In addition, we plan to collect larger and higher-quality healthcare datasets to further enhance prediction accuracy and demonstrate the full potential of our approach in real-world healthcare scenarios.

ACKNOWLEDGMENT

This work is partly supported by NSF under grants 2107449, 2107450, and 2107451.

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